Mock Exam — Weeks 1 to 7

DSA27 · Data Structures and Algorithms · midterm practice · 60 marks: Section A 30 · Section B 30

Question bank practice paper

Fall 2026

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> ***How to use this paper.*** *Sit it in one go, closed book, with no computer. It covers weeks 1–7 — everything before the midterm window — and leans on weeks 4–7, since the weeks 1–3 mock already covers the first three. The questions are taken from the weekly question banks, and* ***none*** *of them repeats the weeks 1–3 mock, so you can sit both. The model answers are in the weekly answers files — look them up* ***only after*** *you have finished and marked yourself.*
>
> *A second paper, mock exam B, prints every question in full and has its own answers. For a new timed paper every time, use the online midterm simulator.*
>
> *The format follows the final:* ***30 marks of multiple choice, 30 marks of written answers.*** *The midterm's own format and date will be announced in the lecture and on the WhatsApp channel.*

Section A — Multiple choice (30 marks)

One mark each. Choose one answer out of four. Write the letters on your answer sheet.

#	Question	#	Question	#	Question
1	W1-M09	11	W3-M11	21	W6-M03
2	W1-M13	12	W3-M19	22	W6-M08
3	W1-M18	13	W4-M03	23	W6-M11
4	W2-M01	14	W4-M07	24	W6-M14
5	W2-M05	15	W4-M10	25	W6-M19
6	W2-M09	16	W4-M14	26	W7-M03
7	W2-M14	17	W4-M18	27	W7-M05
8	W2-M20	18	W5-M04	28	W7-M09
9	W3-M03	19	W5-M09	29	W7-M12

#	Question	#	Question	#	Question
10	W3-M07	20	W5-M17	30	W5-M13

The questions are in the weekly files: week 1 · week 2 · week 3 · week 4 · week 5 · week 6 · week 7.

Section B — Written answers (30 marks)

Answer **all** questions. Marks are shown in brackets.

#	Question	Topic	Marks
B1	W2-E5	the storage rule: what, why, and what remains allowed	4
B2	W2-P2	prove n^2 is not $O(n)$	3
B3	W3-E4	Towers of Hanoi: strategy, recurrence, optimality	4
B4	W4-E1	amortised cost; n appends with doubling are $O(n)$	5
B5	W5-S1	a linked list drawn after every operation	3
B6	W6-E4	shunting-yard, traced on `2 * ( 3 + 4 ) - 5`	4
B7	W7-S1	a ring buffer drawn after every operation	3
B8	W5-K3	which end of a linked list is the front of a queue	2
B9	W6-B3 **and** W7-B3	two bugs, 1 mark each	2
		Total	**30**

Coverage. Section A has three questions from week 1, five from week 2, four from week 3 and 18 from weeks 4–7; Section B gives 11 marks to weeks 2–3 and 19 to weeks 4–7. Searching (week 8) is **not** on this paper; it is on the final.

Marking yourself

- **Section A:** one mark per correct letter. The answer tables are at the top of each weekly answers file.
- **Section B:** use the **marking points** in the answers files. Award a mark for each point your answer makes clearly and correctly — not for length. For B5 and B7, a mark per correct third of the drawing; for B8, one mark for the choice of ends and one for the costs.
- **Then** look at *which* questions you lost marks on, and on which level ([what], [how] or [why]). That tells you what to revise, and how — and the weekly question bank of that week has more of the same kind.